

Companion to Bird & de Moor, Algebra of Programming

§10. Greedy Algorithms

§10.1. Theory

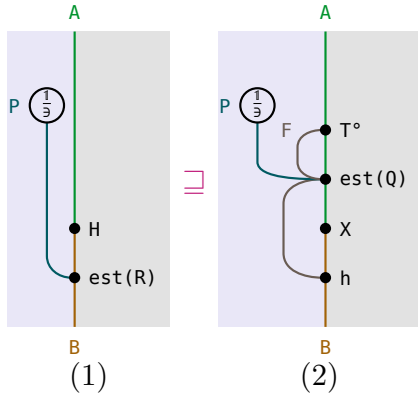
(10.1a)

h, T, R, H, M as in $M \triangleq \frac{H}{\exists} \text{ est}(R)$; additionally Q a **connected** preorder on the sets $\frac{T^\circ}{\exists}$ returns, so that $\frac{T^\circ}{\exists} \text{ est}(Q)$ is entire.

$\text{Map}(h) \wedge h \text{ Pres } R \wedge R \in R \wedge \text{ThinCondition}(T, h, R, Q) \Rightarrow (\mu X : \frac{T^\circ}{\exists} \text{ est}(Q) F(X) h) \sqsubseteq \frac{H}{\exists} \text{ est}(R)$ Theorem 10.1

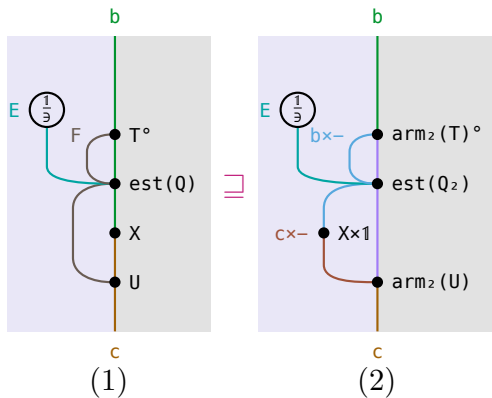
the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried and the recursion runs on values alone

$H \triangleq (T)^\circ (h) \triangleq H, Q$



1	$\frac{H}{\exists} \text{ est}(R)$	
2	$\sqsubseteq \frac{T^\circ}{\exists} \text{ est}(Q) F(X) h$	

the body at T a coproduct of two arms

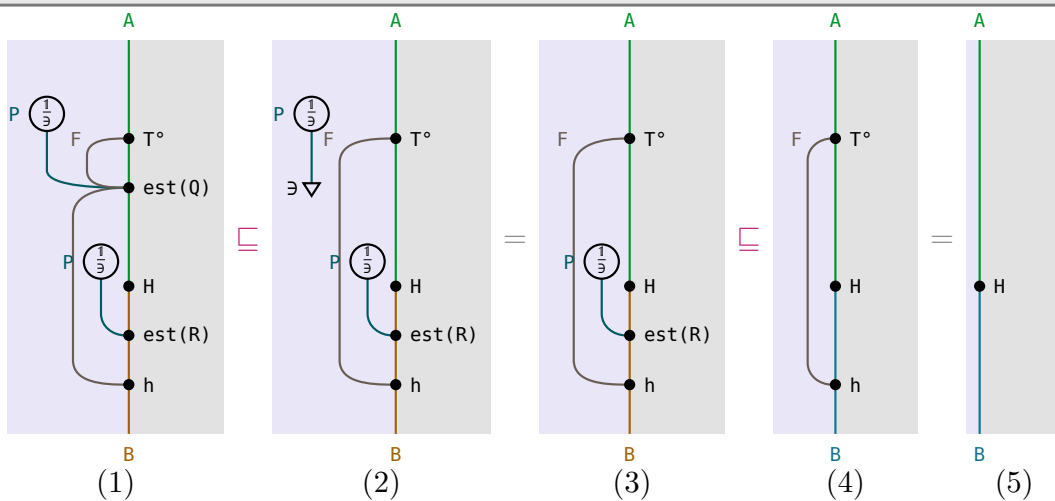


1	$\frac{T^\circ}{\exists} \text{ est}(Q) F(X) U$	Proposition 10.1, $\forall_2 \forall_1^\circ = \emptyset$
2	$\sqsubseteq \frac{\text{arm}_2(T)^\circ}{\exists} \text{ est}(Q_2) (X \times 1) \text{ arm}_2(U)$	

$\frac{T^\circ}{\exists} \text{ est}(Q) F(\frac{H}{\exists} \text{ est}(R)) h \sqsubseteq H$

(10.1b)

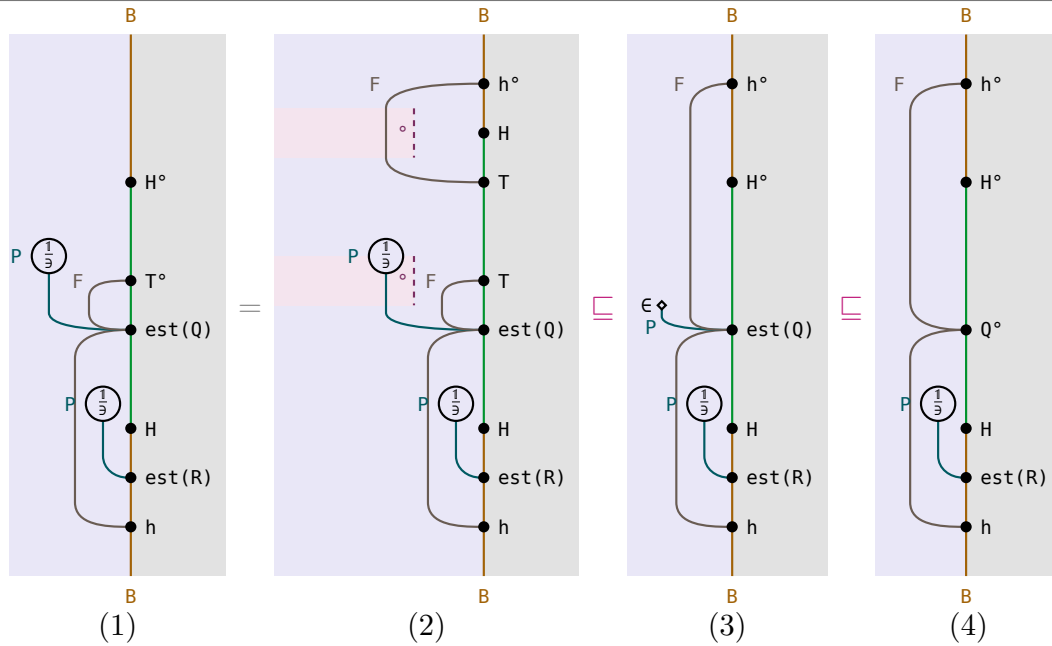
taking the input apart in one Q -extreme way, solving the parts by M and assembling by h returns only what H returns



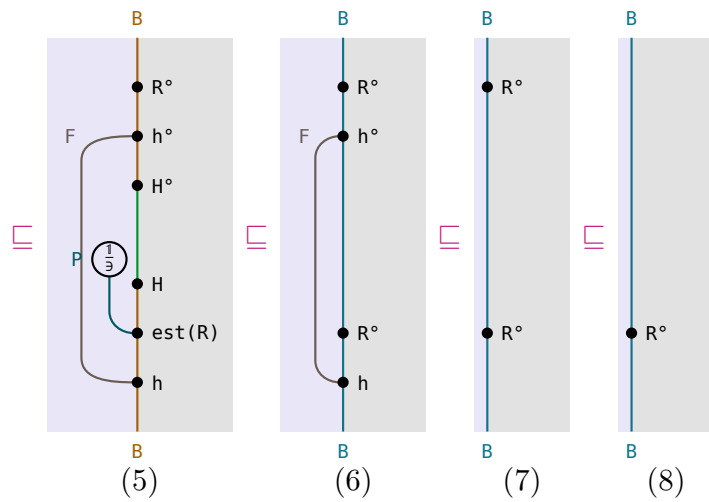
1	$\frac{T^\circ}{\exists} \text{ est}(Q) F(\frac{H}{\exists} \text{ est}(R)) h$	$\text{est}(Q) \sqsubseteq \exists$
2	$\sqsubseteq \frac{T^\circ}{\exists} \exists F(\frac{H}{\exists} \text{ est}(R)) h$	$\frac{R}{\exists} \exists = R$
3	$= T^\circ F(\frac{H}{\exists} \text{ est}(R)) h$	$\frac{\text{est}(R)}{\exists} \sqsubseteq S$
4	$\sqsubseteq T^\circ F(H) h$	$T^\circ F(H) h = H$
5	$= H$	

$$\text{Map}(h) \wedge h \text{ Pres } R \wedge R R \subseteq R \wedge \text{ThinCondition}(T, h, R, Q) \Rightarrow H^\circ \stackrel{T^\circ}{\subseteq} \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h \subseteq R^\circ \quad (10.1c)$$

H° followed by the greedy body at M is $\exists R^\circ$: an answer of the body is never worse than an answer of H to the same input



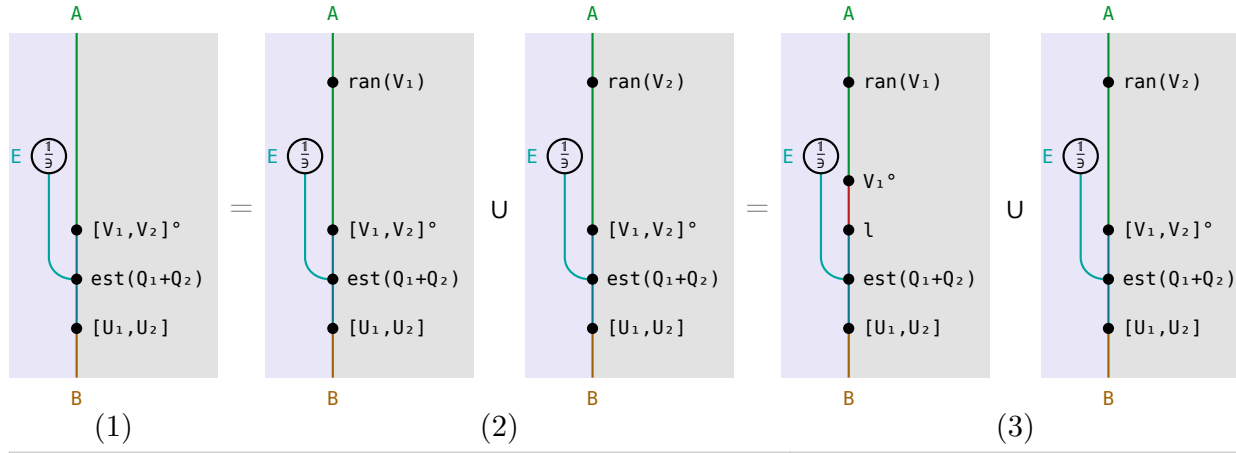
1	$H^\circ \stackrel{T^\circ}{\subseteq} \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h$	$H^\circ = h^\circ F(H^\circ) T$
2	$= h^\circ F(H^\circ) T \stackrel{T^\circ}{\subseteq} \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h$	$T \stackrel{T^\circ}{\subseteq} \epsilon$
3	$\subseteq h^\circ F(H^\circ) \epsilon \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h$	$\epsilon \text{est}(R) \subseteq R^\circ$
4	$\subseteq h^\circ F(H^\circ) Q^\circ F\left(\frac{H}{\exists} \text{est}(R)\right) h$	$\text{ThinCondition}(T, h, R, Q)$ $\Rightarrow h^\circ F(H^\circ) Q^\circ \subseteq R^\circ h^\circ F(H^\circ)$



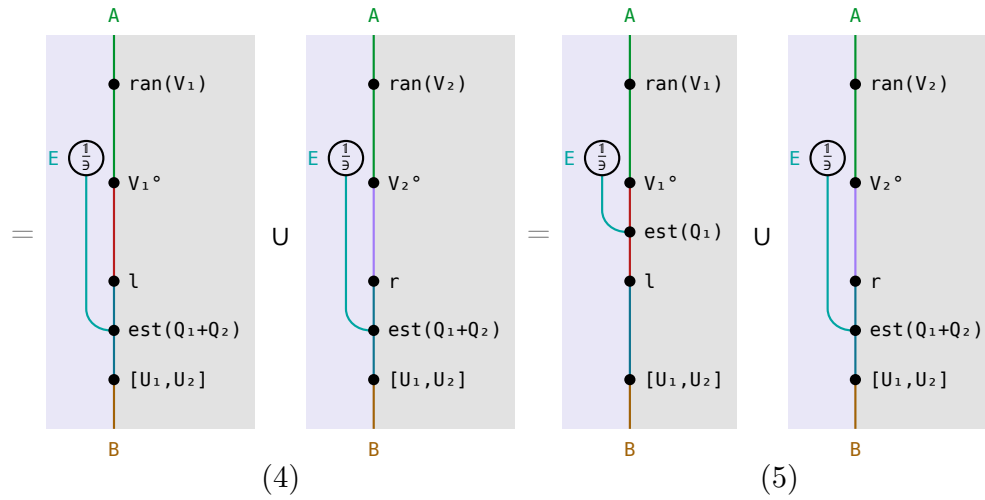
5	$\subseteq R^\circ h^\circ F(H^\circ \frac{H}{\exists} \text{est}(R)) h$	$S^\circ \stackrel{S}{\subseteq} \text{est}(R) \subseteq R^\circ$
6	$\subseteq R^\circ h^\circ F(R^\circ) h$	$\text{Map}(h) \wedge h \text{ Pres } R$ $\Rightarrow h^\circ F(R^\circ) h \subseteq R^\circ$
7	$\subseteq R^\circ R^\circ$	$R R \subseteq R$ $\Rightarrow R^\circ R^\circ \subseteq R^\circ$
8	$\subseteq R^\circ$	

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \ \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} \text{est}(Q_2) U_2) \quad \text{Proposition 10.1}$$

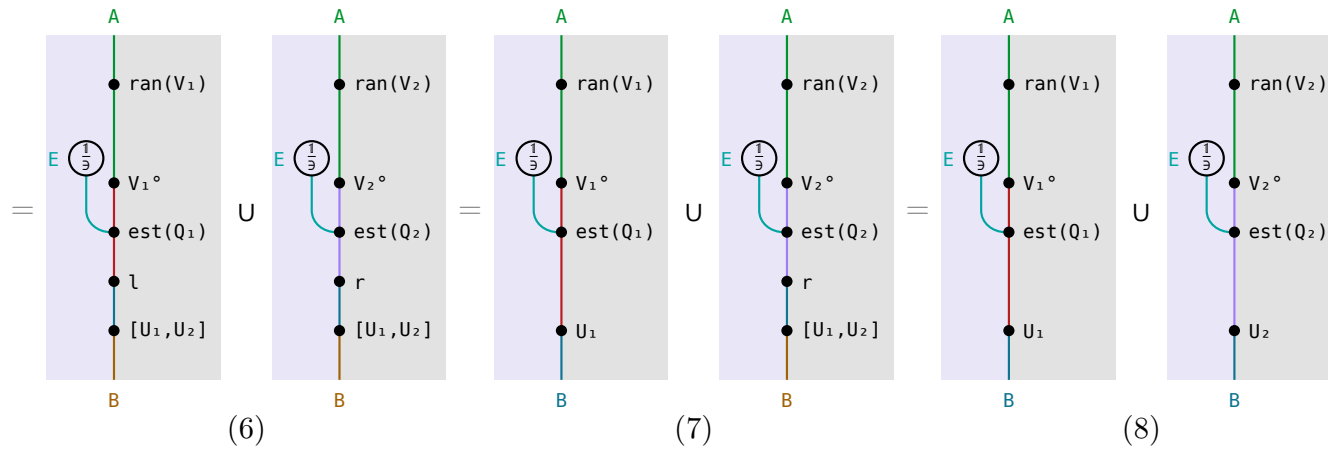
when V_1 and V_2 have disjoint ranges, the greedy step over $[V_1, V_2]$ runs the V_1 step on inputs V_1 reaches and the V_2 step on inputs V_2 reaches



1	$\frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]$	$\frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2])$
2	$= (\text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2])$	$V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{3} = \text{ran}(V_1) \ \frac{V_1^\circ}{3} P(l)$
3	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2])$	$V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{3} = \text{ran}(V_2) \ \frac{V_2^\circ}{3} P(r)$



4	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$	$P(l) \text{est}(Q_1 + Q_2) = \text{est}(Q_1) \ l$
5	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} \text{est}(Q_1) \ l [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$	$P(r) \text{est}(Q_1 + Q_2) = \text{est}(Q_2) \ r$



6	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} \text{est}(Q_1) \ l [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} \text{est}(Q_2) \ r [U_1, U_2])$	$l [R, S] = R$
7	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} \text{est}(Q_2) \ r [U_1, U_2])$	$r [R, S] = S$
8	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} \text{est}(Q_2) U_2)$	

§10.2. The detab-entab problem

§10.2.1. Detab

(10.2.1a)

	name	type	definition	meaning
1	detab	$\text{String} \rightarrow \text{String}$	$\text{detab}([nil, expand]) = \text{detab}$, $\text{detab}(\text{wrap}(a)) = []$, $\text{detab}(\text{snoc}(x, a)) = \text{expand}(\text{detab}(x), a)$	replace every tab by blanks up to the next tab stop; $H = \text{detab}^\circ$
2	$\text{expand}(xs, a)$	String	$\text{expand}(xs, a) \triangleq \text{if } a = \text{tb} \text{ then } \text{fill}(xs) \text{ else } \text{snoc}(xs, a)$	a tab fills the line to the next stop; any other character is appended
3	$\text{fill}(xs)$	String	$\text{fill}(xs) \triangleq xs ++ \text{blanks}(n - \text{col } xs \% n)$	append blanks up to the next multiple of n columns
4	$\text{col}()$	$\text{String} \rightarrow \mathbb{N}$	$\text{col}(\text{wrap}(a)) = 0$, $\text{col}(\text{snoc}(x, a)) = \text{if } a = \text{nl} \text{ then } 0 \text{ else } \text{col}(x) + 1$	the column the string ends at: characters since the last newline
5	R	$[\text{Char}] \rightarrow [\text{Char}]$	$xs \ R \ ys \iff \text{length}(xs) \leq \text{length}(ys)$	x is no longer than y
6	U	$\text{Char} \rightarrow \text{Char}$	$a \ U \ b \iff a = \text{tb} \vee a = b$	a is the tab or $a=b$
7	V	$[\text{Char}] \rightarrow [\text{Char}]$	$xs \ V \ ys \iff \text{prefix}(xs, ys) \wedge \text{fill}(xs) = \text{fill}(ys)$	a prefix that fills to the same string
8	Q	$F[\text{Char}] \rightarrow F[\text{Char}]$	$Q = 1 + (V \times U)$, $l(\text{val}) \ Q \ l(\text{val}') \iff \text{true}$, $r(p) \ Q \ r(q) \iff V(\pi_1(p), \pi_1(q)) \wedge U(\pi_2(p), \pi_2(q))$, $(\forall \text{val}. \forall \text{val}'. u = l(\text{val}) \implies v = l(\text{val}') \implies \text{false}) \wedge (\forall p. \forall q. u = r(p) \implies v = r(q) \implies \text{false}) \implies (u \ Q \ v \iff \text{false})$	compare two decompositions by V on the strings and U on the characters
9	$\text{unfill}()$	$\text{String} \rightarrow \text{String}$	$\text{unfill}(\text{wrap}(a)) = []$, $\text{unfill}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc}(x, a)) \% n \neq 0 \text{ then } \text{unfill}(x) \text{ else } \text{snoc}(x, a)$	the shortest prefix that fills to the same string
10	$\text{tbc}()$	$\text{String} \rightarrow \mathbb{N}$	$\text{tbc}(\text{wrap}(a)) = 0$, $\text{tbc}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc}(x, a)) \% n \neq 0 \text{ then } \text{tbc}(x) + 1 \text{ else } 0$	the number of trailing blanks
11	triple	$[\text{Char}] \rightarrow \text{String} \times (\mathbb{N} \times \mathbb{N})$	$\text{triple} \triangleq (\text{unfill } \text{entab}, (\text{tbc}, \text{col}))$	what entab keeps: the unfilled output, its trailing blank count and its column

§10.2.2. Derivation

$$0 < n \Rightarrow \text{expand } V^\circ \sqsubseteq \text{expand } u (\pi_1 V^\circ)$$

(10.2.2a)

shortening the output of one **expand** step to a V -smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string ($\pi_1 V^\circ$)

(1)	(2)	(3)	(4)
$\text{expand } V^\circ = ((\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } \text{snoc})) V^\circ$		$\text{expand} = (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } \text{snoc})$	
$= (\text{istab } \pi_1 \text{ fill } V^\circ) \cup (\text{nottab } \text{snoc } V^\circ)$		$(S \cup T) R = (S R) \cup (T R)$	
$= (\text{istab } \pi_1 \text{ fill } V^\circ) \cup (\text{nottab } \text{snoc } V^\circ)$		$0 < n \wedge \text{blank} \neq \text{nl} \Rightarrow \text{fill } V^\circ = \text{fill}$	

(5)	(6)	(7)	(8)
$= (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } \text{snoc } V^\circ)$		$0 < n \wedge \text{blank} \neq \text{nl} \Rightarrow \text{snoc } V^\circ \sqsubseteq \text{snoc } u (\pi_1 V^\circ)$	
$\sqsubseteq (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } (\text{snoc } u (\pi_1 V^\circ)))$		$(\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } (\text{snoc } u X)) \sqsubseteq \text{expand} \cup X$	
$\sqsubseteq \text{expand } u (\pi_1 V^\circ)$			

$$0 < n \Rightarrow \text{detab } V^\circ \sqsubseteq R^\circ \text{ detab}$$

(10.2.2b)

any V -shortening of **detab**'s output is **detab**'s output on an input no longer than the given one

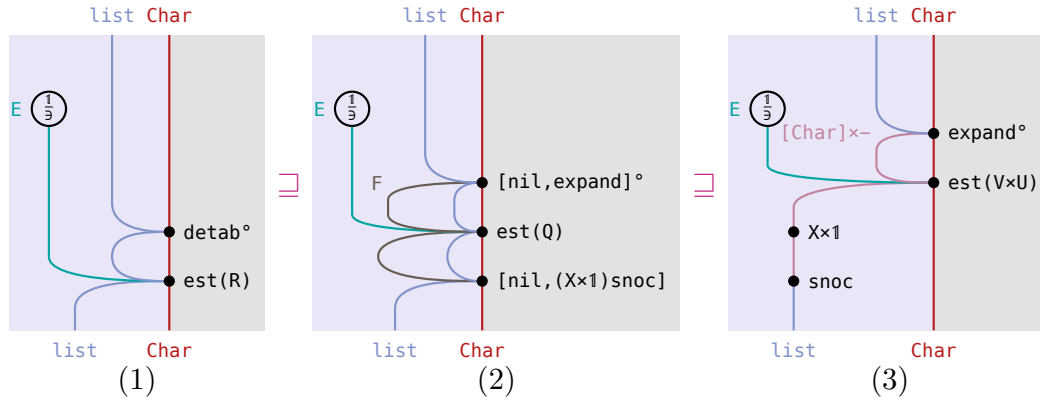
(1)	(2)	(3)	(4)
$\text{detab } V^\circ = [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}] V^\circ$		$\text{detab} = [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}]$	
$= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}] V^\circ$		$[R, S] Z = [R Z, S Z]$	
$= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil } V^\circ, \text{expand } V^\circ]$		$\text{nil } V^\circ = \text{nil}$	
$= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } V^\circ]$		$(10.2.2a) 0 < n \Rightarrow \text{expand } V^\circ \sqsubseteq \text{expand } u (\pi_1 V^\circ)$	

(5)	(6)	(7)	(8)
$\sqsubseteq [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } u (\pi_1 V^\circ)]$		$[\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } u X] = \text{detab } u (\text{snoc}^\circ (\text{detab } \times 1) X)$	
$= \text{detab } u (\text{snoc}^\circ (\text{detab } \times 1) \pi_1 V^\circ)$		$\text{snoc}^\circ (\text{detab } \times 1) \pi_1 Y = \text{snoc}^\circ \pi_1 \text{ detab } Y$	
$= \text{detab } u (\text{snoc}^\circ \pi_1 \text{ detab } V^\circ)$		$0 < n \Rightarrow \text{detab } u (\text{snoc}^\circ \pi_1 \text{ detab } V^\circ) \sqsubseteq R^\circ \text{ detab}$	
$\sqsubseteq R^\circ \text{ detab}$			

$$0 < n \Rightarrow (\mu X : \frac{[nil, expand]^{\circ}}{3} \text{ est}(Q) F(X) [nil, snoc]) \sqsubseteq \frac{detab^{\circ}}{3} \text{ est}(R) \quad (10.2.2c)$$

the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and cashing the held blanks in for a tab wherever the column reaches a tab stop

the specification — $detab(wrap(a)) = []$, $detab(snoc(x,a)) = expand(detab(x),a)$; $detab \text{ entab} = 1$ and nothing shorter does



$$1 \quad \frac{detab^{\circ}}{3} \text{ est}(R)$$

at $Q \triangleq 1 + (V \times U)$: one character of input is decided at each step. $F(\tau, R) \alpha \sqsubseteq \alpha R$. $detab \text{ prefix} \sqsubseteq R^{\circ} \text{ detab}$ is FALSE, — at $n=8$, $detab [a,b,c,d,e,TB] = [a,b,c,d,e,BL,BL,BL]$, whose prefix $[a,b,c,d,e,BL,BL]$ is longer than any input giving it, and $detab V^{\circ} \sqsubseteq R^{\circ} \text{ detab}$ holds. $expand V^{\circ} \sqsubseteq expand \cup (\pi_1 V^{\circ})$ — shortening the output either leaves the last step alone or discards it

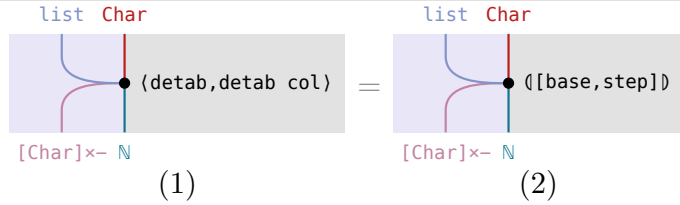
$$2 \quad \sqsubseteq \frac{[nil, expand]^{\circ}}{3} \text{ est}(Q) F(X) [nil, snoc]$$

Proposition 10.1: **nil** and **expand** have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop

$$3 \quad \sqsubseteq \frac{expand^{\circ}}{3} \text{ est}(V \times U) (X \times 1) snoc$$

$$\langle detab, detab \text{ col} \rangle = \langle [base, step] \rangle \quad (10.2.2d)$$

one pass over the input carries the output so far together with its column



$$1 \quad \langle detab, detab \text{ col} \rangle$$

both components are folds over the same input; **step** on $(detab \ x, col(detab \ x))$ is $(detab, col \ detab)$ one character further — not a tabulated row

$$2 \quad = \langle [base, step] \rangle$$

Exercise 10.1

$$\langle [base, step] \rangle (convert(xs), r) \Leftrightarrow r = loop(step, ([], 0), xs)$$

the fold of $[base, step]$ over the input converted to a snoc-list is the left loop of **step** from **base**

$$\pi_1(loop(step, ([], c_0), xs)) = loop'((f, g), (c_0, xs))$$

Exercise 10.1: when **step** only appends $f(c, a)$ to the output and moves the state by g , the output of the loop is $loop'(f, g)$, which never carries the output

$$a \neq tb \Rightarrow expand(ys, b) = snoc(xs, a) \Leftrightarrow fill(ys) = snoc(xs, a) \wedge b = tb \vee (ys, b) = (xs, a) \quad (10.2.2e)$$

expand produces $x\# [a]$ from a tab after a string that fills to it, or from x and a

$$0 < n \Rightarrow (\exists ys. fill(ys) = snoc(xs, a)) \Leftrightarrow a = blank \wedge col(snoc(xs, a)) \% n = 0$$

a string is a **fill** exactly when it ends in a blank on a tab stop

$$entab(xs) = entab(unfill(xs)) ++ blanks(tbc(xs)) \quad (10.1)$$

(10.1): the output of **entab** is that of **unfill** x followed by the trailing blanks held back

$$tbc[] = 0$$

the empty string has no trailing blanks

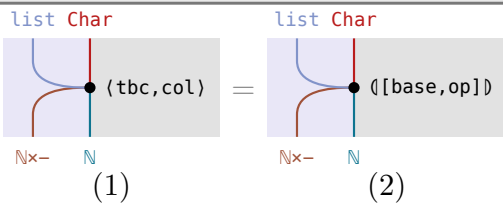
$$tbc(snoc(xs, a)) = \text{if } a = \text{blank} \wedge col(snoc(xs, a)) \% n \neq 0 \text{ then } tbc(xs) + 1 \text{ else } 0$$

a blank off a tab stop adds one held blank; anything else releases them all

(tbc,col) = ([base,op])

(10.2.2f)

one pass counts the held blanks and the column together



1 (tbc,col)

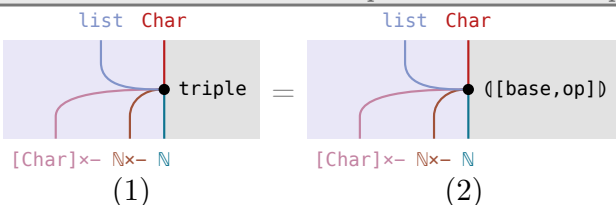
2 = ([base,op])

op on (tbc x,col x) is (tbc,col) one character further,
 from the equations for tbc — (10.1) entab(xs) =
 entab(unfill(xs)) ++ blanks(tbc(xs)), tbc[] = 0,
 tbc(snoc(xs,a)) = if a = blank $\wedge$
 col(snoc(xs,a)) % n $\neq$ 0 then tbc(xs) + 1 else 0

triple = ([base,op])

(10.2.2g)

one pass carries the output up to the held blanks, their count and the column



1 triple

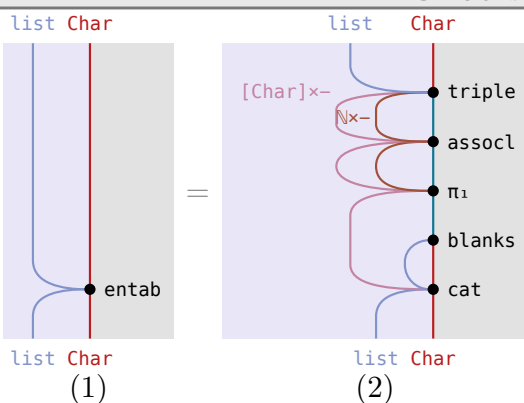
2 = ([base,op])

op on triple x is triple one character further: (10.1) where
 the blanks are released — (10.1) entab(xs) =
 entab(unfill(xs)) ++ blanks(tbc(xs)), tbc[] = 0,
 tbc(snoc(xs,a)) = if a = blank $\wedge$
 col(snoc(xs,a)) % n $\neq$ 0 then tbc(xs) + 1 else 0

entab = triple assocl π_1 (1 $\times$ blanks) cat

(10.2.2h)

entab is triple with the held blanks appended to its output



1 entab

2 = triple assocl π_1 (1 $\times$ blanks) cat

(10.1) — (10.1) entab(xs) = entab(unfill(xs)) ++
 blanks(tbc(xs)), tbc[] = 0, tbc(snoc(xs,a)) = if
 a = blank $\wedge$ col(snoc(xs,a)) % n $\neq$ 0 then
 tbc(xs) + 1 else 0

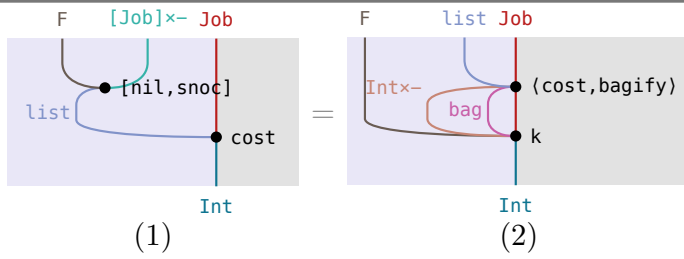
§10.3. The minimum tardiness problem

(10.3a)

	name	type	definition	meaning
1	bag(Job)	$\mathcal{A}$	$\text{bag}(\text{Job}) \triangleq \text{Quotient}(\text{perm})$	a bag of jobs: a list up to reordering
2	snag(p)	bag(Job)	$\text{snag}(p) \triangleq \pi_2(p) :: \pi_1(p)$	put one job into a bag
3	bagify	[Job]→bag(Job)	$\langle [\text{nil}, \text{snag}] \rangle = \text{bagify}, \text{bagify}(\text{wrap}(u)) = \text{nil},$ $\text{bagify}(\text{snoc}(xs, a)) = \text{snag}(\text{bagify}(xs), a)$	forget the order of a schedule; $H = \text{bagify}^\circ$ lists every schedule of a bag
4	penalty(xs, j)	Int	$\text{penalty}(xs, j) \triangleq (\text{sum}(\text{list}(\text{ct}))(xs) + \text{ct}(j) - \text{dt}(j)) \text{wt}(j)$	how late job j finishes after the schedule xs, times its weight
5	cost	[Job]→Int	$\text{cost}(\text{wrap}(a)) = 0, \text{cost}(\text{snoc}(x, a)) =$ $\text{bmax}(\text{cost}(x), \text{penalty}(x, a))$	the largest penalty of any job in the schedule
6	bmax(a, b)	Int	$\text{bmax}(a, b) \triangleq \text{if } a \leq b \text{ then } b \text{ else } a$	the larger of two numbers
7	R	[Job]→[Job]	$R = \text{cost} \leq \text{cost}^\circ, R(xs, ys) \iff \text{cost}(xs) \leq \text{cost}(ys)$	xs costs no more than ys
8	add	[Job]×Job→[Job]	$p \text{ add } ys \iff \text{add}(\pi_2(p), ys)$	insert j anywhere in xs; $\text{perm} \triangleq \text{bagify}$ $\text{bagify}^\circ = \langle [\text{nil}, \text{add}] \rangle$
9	k	F(Int×bag(Job))→Int	$k \triangleq [\text{nil}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}],$ $k(l(u)) = 0, k(r((c, xs), j)) =$ $\text{bmax}(c, (\text{bagify}^\circ \times 1) \text{ penalty}((xs, j)))$	the cost of a schedule from the cost of its front and the penalty of its last job
10	f	F(bag(Job))→Int	$f(l(\text{val})) = 0, f(r(p)) = (\text{bagify}^\circ \times 1) \text{ penalty}(p)$	the penalty of the last job, read off the bag before it
11	Q	F(bag(Job))→F(bag(Job))	$u Q v \iff f(u) \leq f(v)$	the last job of u gets no larger penalty than that of v
12	Q'	bag(Job)×Job→bag(Job)×Job	$p Q' q \iff (\text{bagify}^\circ \times 1) \text{ penalty}(p) \leq$ $(\text{bagify}^\circ \times 1) \text{ penalty}(q)$	Q on the snag summand: compare the penalties of the two last jobs
13	g	F[Job]→Int	$g \triangleq [\text{zero}, \text{penalty}], g(s, c) \iff c = g(s), g(l(u)) = 0,$ $g(r(xs, j)) = \text{penalty}(xs, j)$	(10.5): the penalty of the last job, zero for the empty schedule
14	m	F[Job]→Int	$m \triangleq [\text{zero}, \pi_1 \text{ cost}]$	(10.6), B&dM's h renamed as in (Vj. $0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j))$ $\implies (\mu X : \frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q)$ $F(X) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{bagify}^\circ}{\exists} \text{ est}(R) :$ the cost of the schedule before the last job

$$[\text{nil}, \text{snoc}] \text{ cost} = F(\langle \text{cost}, \text{bagify} \rangle) k \quad (10.3b)$$

the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job



1 $[\text{nil}, \text{snoc}] \text{ cost}$

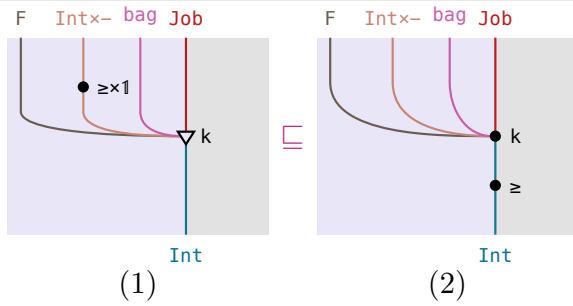
$\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$, and
 $\text{penalty}(xs, j)$ reads only the bag of xs — $\text{penalty}(xs, j) \triangleq (\text{sum}(\text{list}(\text{ct}))(xs) + \text{ct}(j) - \text{dt}(j)) \text{wt}(j)$

2 $= F(\langle \text{cost}, \text{bagify} \rangle) k$

$$F(\geq \times 1) k \sqsubseteq k \geq$$

Exercise 10.5

a larger cost before the last job gives a larger cost after it



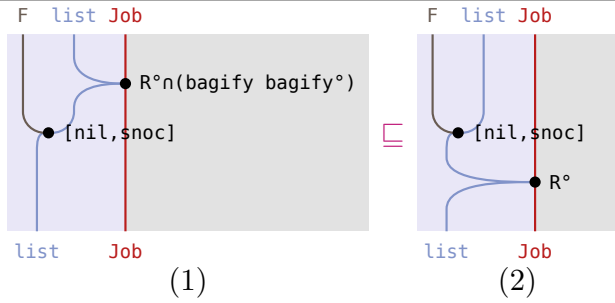
1 $F(\geq \times 1) k$

bmax is monotone in its first argument

2 $\sqsubseteq k \geq$

$$F(R^\circ \circ n(\text{bagify } \text{bagify}^\circ)) [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R^\circ \quad (10.2)$$

improving the schedule before the last job, among schedules of the same bag, improves the whole schedule



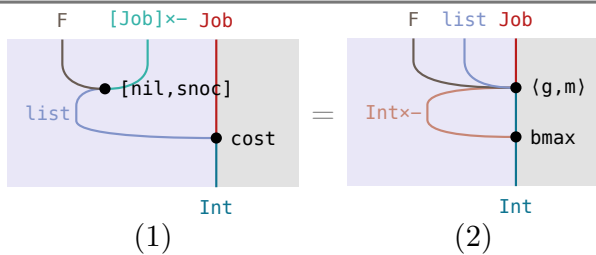
1 $F(R^\circ \circ n(\text{bagify } \text{bagify}^\circ)) [\text{nil}, \text{snoc}]$

Proposition 9.3 at $S \triangleq \text{bagify}$, $\leq \triangleq$ — Proposition 9.3
 $\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge h \text{ cost} = F(\langle \text{cost}, S \rangle) k$
 $\wedge F(\leq \times 1) k \sqsubseteq k \leq$
 $\Rightarrow F(\langle \text{cost}^\circ, S^\circ \rangle \circ n(S^\circ)) h \sqsubseteq h \text{ cost} \leq$
 cost° , with $[\text{nil}, \text{snoc}] \text{ cost} = F(\langle \text{cost}, \text{bagify} \rangle) k$
 and Exercise 10.5 $F(\geq \times 1) k \sqsubseteq k \geq$

2 $\sqsubseteq [\text{nil}, \text{snoc}] R^\circ$

$$[\text{nil}, \text{snoc}] \text{ cost} = \langle g, m \rangle \text{ bmax} \quad (10.4)$$

the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it



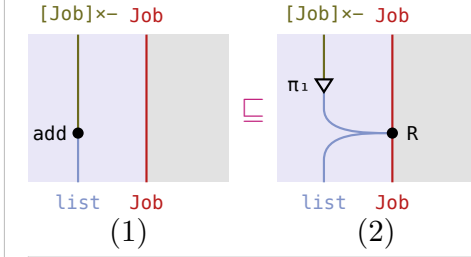
1 $[\text{nil}, \text{snoc}] \text{ cost}$

$\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$ and
 bmax commutes — $\text{cost}(\text{wrap}(a)) = 0$,
 $\text{cost}(\text{snoc}(x, a)) = \text{bmax}(\text{cost}(x), \text{penalty}(x, a))$

2 $= \langle g, m \rangle \text{ bmax}$

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow add \sqsubseteq \pi_1 R \quad (10.7)$$

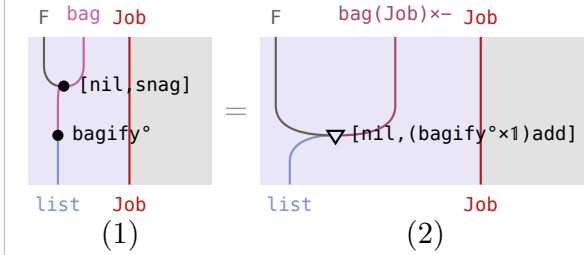
putting one more job anywhere into a schedule never lowers its cost



1	add	induction on where j goes: every job after it starts later, so no penalty falls
2	$\sqsubseteq \pi_1 R$	

$$[nil, snag] \text{ bagify}^\circ = [nil, (\text{bagify}^\circ \times 1) \text{ add}] \quad (10.8)$$

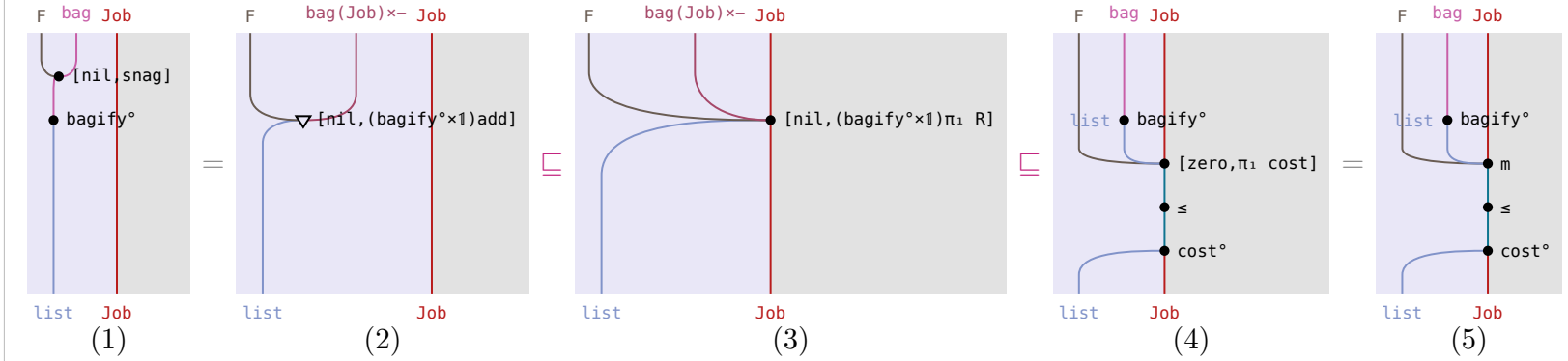
the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere



1	$[nil, snag] \text{ bagify}^\circ$	take out the last copy of j and put it back — $p \text{ add } ys \Leftrightarrow \text{add}(\pi_2(p), ys)$
2	$= [nil, (\text{bagify}^\circ \times 1) \text{ add}]$	

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow [nil, snag] \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) m \leq \text{cost}^\circ \quad (10.3c)$$

an ordering of a bag with one more job costs at least the cost of the ordering of the bag before its last job



1	$[nil, snag] \text{ bagify}^\circ$	(10.8) $[nil, snag] \text{ bagify}^\circ = [nil, (\text{bagify}^\circ \times 1) \text{ add}]$
2	$= [nil, (\text{bagify}^\circ \times 1) \text{ add}]$	(10.7) $(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow add \sqsubseteq \pi_1 R$
3	$\sqsubseteq [nil, (\text{bagify}^\circ \times 1) \pi_1 R]$	$[nil, \pi_1 R] \sqsubseteq [zero, \pi_1 \text{ cost}] \leq \text{cost}^\circ$
4	$\sqsubseteq F(\text{bagify}^\circ) [zero, \pi_1 \text{ cost}] \leq \text{cost}^\circ$	
5	$= F(\text{bagify}^\circ) m \leq \text{cost}^\circ$	

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow (Q \cap ([nil, snag] [nil, snag]^\circ)) F(bagify^\circ) [nil, snoc] \sqsubseteq F(bagify^\circ) [nil, snoc] R \quad (10.3)$$

a schedule of a bag ending in a job of least penalty is no worse than any schedule of the same bag

(1)	(2)	(3)	(4)
1 $(Q \cap ([nil, snag] [nil, snag]^\circ)) F(bagify^\circ) [nil, snoc]$	$(S \cap T) R \sqsubseteq (S R) \cap (T R)$		
2 $\sqsubseteq Q F(bagify^\circ) [nil, snoc] \cap ([nil, snag] [nil, snag]^\circ F(bagify^\circ) [nil, snoc])$	$[nil, snag]^\circ F(bagify^\circ) [nil, snoc] = bagify^\circ$		
3 $= (Q F(bagify^\circ) [nil, snoc]) \cap ([nil, snag] bagify^\circ)$	(10.3c) $(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j))$		
	$\Rightarrow [nil, snag] bagify^\circ \sqsubseteq F(bagify^\circ) m \leq cost^\circ$		
4 $\sqsubseteq (Q F(bagify^\circ) [nil, snoc]) \cap (F(bagify^\circ) m \leq cost^\circ)$	$(Q F(bagify^\circ) [nil, snoc]) \cap (F(bagify^\circ) M) \sqsubseteq F(bagify^\circ) ((F(bagify^\circ) Q F(bagify^\circ)) \cap (M [nil, snoc]^\circ)) [nil, snoc]$		
(5)	(6)	(7)	(8)
5 $\sqsubseteq F(bagify^\circ) ((F(bagify^\circ) Q F(bagify^\circ)) \cap (m \leq cost^\circ [nil, snoc]^\circ)) [nil, snoc]$	$F(bagify^\circ) Q F(bagify^\circ) = g \leq g^\circ$		
6 $= F(bagify^\circ) ((g \leq g^\circ) \cap (m \leq cost^\circ [nil, snoc]^\circ)) [nil, snoc]$	$(R S)^\circ = S^\circ R^\circ$		
7 $= F(bagify^\circ) ((g \leq g^\circ) \cap (m \leq ([nil, snoc] cost)^\circ)) [nil, snoc]$	$\langle X, Y \rangle \langle R, S \rangle^\circ = \langle X R^\circ \rangle \cap \langle Y S^\circ \rangle$		
8 $= F(bagify^\circ) \langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ [nil, snoc]$	(10.3d) $\langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ$		
9 $\sqsubseteq F(bagify^\circ) [nil, snoc] R$	$[nil, snoc] \sqsubseteq [nil, snoc] R$		

$$\langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ [nil, snoc] \sqsubseteq [nil, snoc] R \quad (10.3d)$$

a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job, gives a schedule costing at most the whole

(1) (2) (3)

1	$\langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ [nil, snoc]$	$1 \sqsubseteq cost cost^\circ$
2	$\sqsubseteq \langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ [nil, snoc] cost cost^\circ$	$[nil, snoc] cost = \langle g, [nil, snoc] cost \rangle bmax$
3	$= \langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ \langle g, [nil, snoc] cost \rangle bmax cost^\circ$	$\langle g, [nil, snoc] cost \rangle^\circ \langle g, [nil, snoc] cost \rangle \sqsubseteq 1$

(4) (5) (6) (7)

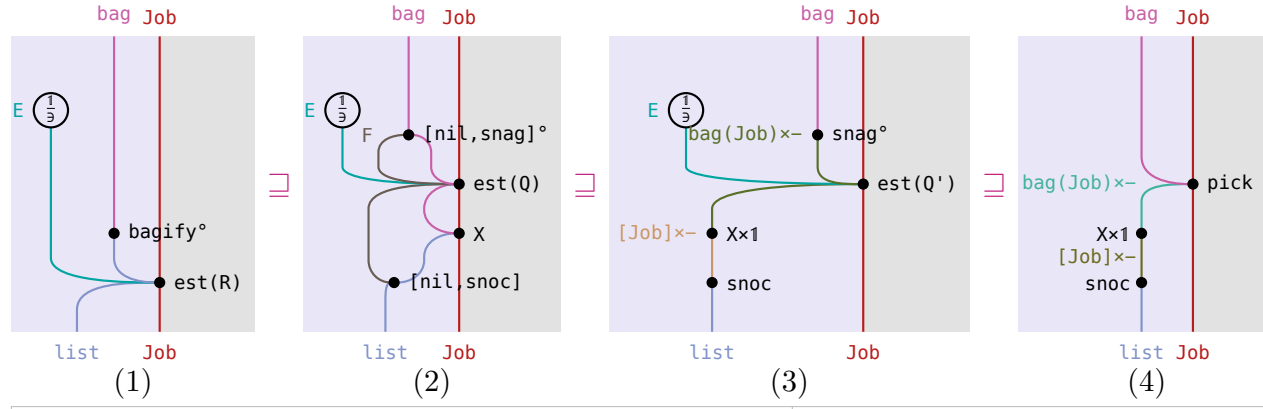
4	$\sqsubseteq \langle g \leq, m \leq \rangle bmax cost^\circ$	$\langle g \leq, m \leq \rangle bmax \sqsubseteq \langle g, m \rangle bmax \leq$
5	$\sqsubseteq \langle g, m \rangle bmax \leq cost^\circ$	(10.4) $[nil, snoc] cost = \langle g, m \rangle bmax$
6	$= [nil, snoc] cost \leq cost^\circ$	$R = cost \leq cost^\circ$
7	$= [nil, snoc] R$	

$$\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick} (\text{schedule} \times 1) \text{ snoc}), (\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \quad (10.3e)$$

$$\frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) \text{ F}(X) [\text{nil}, \text{snoc}] \sqsubseteq \frac{\text{bagify}^\circ}{\exists} \text{ est}(R)$$

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag, putting it last, and scheduling what is left the same way

the specification — $\text{bagify} \triangleq \text{bagify}$



1	$\frac{\text{bagify}^\circ}{\exists} \text{ est}(R)$	No greedy reduce exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole
2	$\sqsubseteq \frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) \text{ F}(X) [\text{nil}, \text{snoc}]$	Proposition 10.1: nil and snag have disjoint ranges
3	$\sqsubseteq \frac{\text{snag}^\circ}{\exists} \text{ est}(Q') (X \times 1) \text{ snoc}$	$\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{\exists} \text{ est}(Q')$, a partial function, quadratic in the number of jobs
4	$\sqsubseteq \text{pick} (X \times 1) \text{ snoc}$	

§10.4. The TeX problem

(10.4a)

	name	type	definition	meaning
1	intern	$[\text{Digit}] \rightarrow [0, 2^{16})$	$\text{intern} \triangleq \text{val round}$	the value of a decimal fraction, rounded to 16 binary places
2	val	$[\text{Digit}] \rightarrow \text{Real}$	$\text{val} \triangleq ([\text{zero}, \text{shift}])$	the real number a digit list denotes after the decimal point
3	shift	$\text{Digit} \times \text{Real} \rightarrow \text{Real}$	$\text{shift} \triangleq \text{shift}, \text{shift}((d, r)) \triangleq (d + r)/10$	put digit d in front of r: $(d+r)/10$
4	round	$\text{Real} \rightarrow [0, 2^{16})$	$r \text{ round } n \iff (2n-1)/w < r \wedge r < (2n+1)/w, \text{round}^\circ = \text{interval inrange}$	the nearest integer to $2^{16}r$: $\text{round } r = n$ iff $2n-1 < 2^{17}r < 2n+1$
5	interval	$[0, 2^{16}) \rightarrow \text{Interval}$	$\text{interval}(n) \triangleq ((2n-1)/w, (2n+1)/w)$	the open interval $((2n-1)/2^{17}, (2n+1)/2^{17})$ of the reals that round to n
6	inrange	$\text{Interval} \rightarrow \text{Real}$	$p \text{ inrange } r \iff \text{lo}(p) < r \wedge r < \text{hi}(p)$	r lies strictly inside (a,b)
7	Legal(p)	Prop	$\text{Legal}(p) \triangleq 0 < \text{hi}(p) \wedge \text{hi}(p) < 1 \wedge \text{lo}(p) < \text{hi}(p)$	(10.9): $0 < b < 1$ and $a < b$; Interval is the legal pairs
8	arb	$1 \rightarrow \text{Interval}$	$x \text{ arb } p \iff \text{lo}(p) < 0 \wedge 0 < \text{hi}(p)$	any legal interval
9	step	$\text{Digit} \times \text{Interval} \rightarrow \text{Interval}$	$\text{step}(p) \triangleq (\text{shift}((\pi_1(p), \text{lo}(\pi_2(p)))), \text{shift}((\pi_1(p), \text{hi}(\pi_2(p)))))$	put digit d in front of both ends: $((d+a)/10, (d+b)/10)$
10	H	$\text{Interval} \rightarrow [\text{Digit}]$	$H \triangleq ([\text{arb}, \text{step}])^\circ$	every digit list whose interval the given one is
11	R	$[\text{Digit}] \rightarrow [\text{Digit}]$	$xs \text{ R } ys \iff \text{length}(xs) \leq \text{length}(ys)$	x is no longer than y
12	l	$1 \rightarrow \text{FX}$	$l \triangleq \text{inl}$	the injection of the empty case into $\text{FX} = 1 + (\text{Digit} \times X)$
13	r	$\text{Digit} \times X \rightarrow \text{FX}$	$r \triangleq \text{inr}$	the injection of the digit case into $\text{FX} = 1 + (\text{Digit} \times X)$
14	!	$\text{Digit} \times \text{Interval} \rightarrow 1$	$! \triangleq \rightarrow ()$	the one map to 1
15	Q	$F(\text{Interval}) \rightarrow F(\text{Interval})$	$Q \triangleq (l^\circ !^\circ r) \cup 1, Q(u, z) \iff (\exists t. u = l(t) \wedge (\exists s. z = r(s))) \vee u = z$	u is the empty case and z a digit case, or $u=z$
16	w	Int	$w \triangleq 131072$	2^{17}

$([arb, step]) \sqsubseteq \text{val inrange}^\circ$

(10.4b)

every interval got by folding a decimal's digits with $[arb, step]$ has the decimal's value strictly inside it

1	$F(\text{inrange}^\circ) [arb, step]$	$F(S) [T, U] = [T, (1 \times S) U]$
2	$= [arb, (1 \times \text{inrange}^\circ) step]$	$(1 \times \text{inrange}^\circ) step \sqsubseteq \text{shift inrange}^\circ$
3	$\sqsubseteq [arb, \text{shift inrange}^\circ]$	$\text{zero inrange}^\circ = arb$
4	$= [\text{zero inrange}^\circ, \text{shift inrange}^\circ]$	$[R, S] Z = [R Z, S Z]$
5	$= [\text{zero}, \text{shift}] \text{inrange}^\circ$	

 $Q F(X) \alpha \sqsubseteq F(X) \alpha R$

(10.4c)

choosing by Q among the one-step decompositions before building with $F(X)$ and α yields only decimals that are R -related to one built without choosing

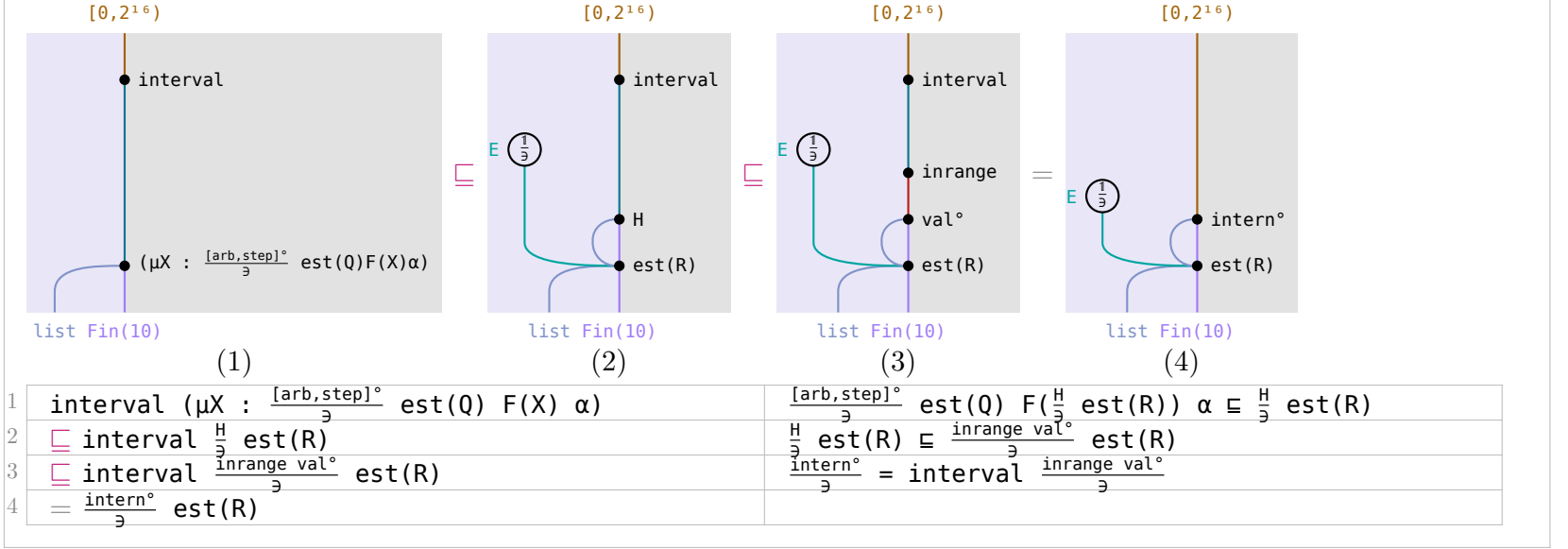
1	$Q F(X) \alpha$	$Q \triangleq (\text{!}^\circ \text{!}^\circ r) \cup 1$
2	$\triangleq ((\text{!}^\circ \text{!}^\circ r) \cup 1) F(X) \alpha$	$(S \cup T) R = (S R) \cup (T R)$
3	$= (\text{!}^\circ \text{!}^\circ r F(X) \alpha) \cup (F(X) \alpha)$	$1 \sqsubseteq R$

4	$\sqsubseteq (\text{!}^\circ \text{!}^\circ r F(X) \alpha) \cup (F(X) \alpha R)$	$r F(X) \alpha \sqsubseteq \text{!} \text{!} \alpha R$
5	$\sqsubseteq (\text{!}^\circ \text{!}^\circ \text{!} \alpha R) \cup (F(X) \alpha R)$	$\text{!}^\circ \text{!} \sqsubseteq 1$
6	$\sqsubseteq (\text{!}^\circ \text{!} \alpha R) \cup (F(X) \alpha R)$	$\text{!} F(X) = \text{!}$

7	$= (\text{!}^\circ \text{!} F(X) \alpha R) \cup (F(X) \alpha R)$	$\text{!}^\circ \text{!} \sqsubseteq 1$
8	$\sqsubseteq (F(X) \alpha R) \cup (F(X) \alpha R)$	$R \cup R = R$
9	$= F(X) \alpha R$	

$$\text{extern}(n)=f(2n-1,2n+1), \quad \text{interval } (\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{ est}(R) \quad (10.4d)$$

a shortest decimal whose internal representation is the given multiple of 2^{-16} is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do



- $f(a,b) = xs \iff a < 0 \wedge xs = [] \vee \neg a < 0 \wedge (\exists d. \exists ys. d = [10b] \wedge f(10a-d, 10b-d) = ys \wedge xs = [d] \# ys)$
 f gives (a,b) the empty decimal when $a < 0$, and otherwise the digit $d = [10b]$ followed by what it gives $(10a-d, 10b-d)$ (10.4e)
- $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ est}(Q) F(X) \alpha) = f$
the least solution of the greedy recursion of $\text{interval } (\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{ est}(R)$ is f
- $d = [10b] \wedge e = [10b] \implies d = e$
Exercise 10.12: two digits d and e with $0 < 10b-d < 1$ and $0 < 10b-e < 1$ are equal
- $(0 < q-p \wedge q < w) \wedge p < 0 \implies f(p,q) = []$
the program's first clause: $f(p,q) = []$ when $p < 0$
- $(0 < q-p \wedge q < w) \wedge \neg p < 0 \implies f(p,q) = [10q/w] \# f(10p-w \cdot (10q/w), 10q-w \cdot (10q/w))$
the program's second clause: with $d = (10q) \text{ div } w$, $f(p,q)$ is d followed by $f(10p-w \cdot d, 10q-w \cdot d)$
- $(0 < q-p \wedge q < w) \wedge \text{Reach}(p,q) \implies f(p/w, q/w) = f(p,q)$
on the pairs the program reaches from $\text{interval } n$, the integer f at (p,q) is a value of the rational f at $(p/w, q/w)$
- $\text{interval } f = \text{extern}$
 f after interval is the program extern , whose value at n is $f(2n-1, 2n+1)$

§10. Appendix

§10.1. $P(S) \text{ est}(R) = (\exists S) \cap (\epsilon \setminus (SR^\circ))$

(10.1a)

(p, q) tabulates $W \triangleq (\exists S) \cap (\epsilon \setminus (SR^\circ))$, and $y \triangleq \frac{(p \exists S) \cap (qR)}{\exists}$, a map.

(10.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq y y^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(10.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f^\circ, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) \cap (qR) \end{array}} \xleftarrow{\Delta \neg \cap, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

$$\boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, ^\circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple}$$

(10.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg \cap, \cdot T \neg / T, ^\circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array}$$

$$\boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple}$$

$$\boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) \cap (qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq q q^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{^\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half}$$

§10.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$

(10.2a)

$$\boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)}$$

$$\begin{array}{ccc} \xrightarrow{\text{P a relator}} \boxed{P(\text{dom}(\text{est}(R)))P(\text{est}(R)) \text{ est}(R)} & \xrightarrow{(7.11) \ R \ R \sqsubseteq R} & \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \\ \Rightarrow P(\text{est}(R)) \text{ est}(R) \sqsubseteq \text{union est}(R) & & \end{array}$$

(10.2b)

$$\begin{array}{ccc} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} & \xleftarrow{\text{P}(\text{dom}(\text{est}(R))) \text{ union est}(R) \sqsubseteq \text{est}(R)} & \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \\ \boxed{P(\text{est}(R)) \text{ est}(R)} & \xRightarrow{\substack{1 \sqsubseteq R \\ \Rightarrow P(S) \text{ est}(R) = (\exists S) \cap (\epsilon \setminus (S R^\circ)) \text{ at } S := \text{est}(R), \\ \Delta \vdash n, T \vdash \neg T \setminus}} & \boxed{\epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \\ & & \boxed{\text{est}(R) R^\circ} \end{array}$$

$$\boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \xleftarrow{P(\text{dom}(\text{est}(R))) \sqsubseteq 1} \boxed{\text{union est}(R)}$$

$$\begin{array}{ccc} \xleftarrow{T(\text{UnV}) \sqsubseteq T \text{UnTV}, \cdot \exists \vdash \frac{}{\exists}, (7.3) \text{ union } (\epsilon \setminus R) = \epsilon \setminus (\epsilon \setminus R)} & \boxed{(\exists \exists) n (\epsilon \setminus (\epsilon \setminus R^\circ))} & \xleftarrow{\text{modular law}} \boxed{\exists (\exists n (\epsilon \setminus (\epsilon \setminus R^\circ)))} \\ & \boxed{\exists \text{est}(R)} & \boxed{\exists (\exists n (\epsilon \setminus R^\circ))} \\ & & \text{count of } T \cdot \neg T \setminus \end{array}$$

$$\boxed{\epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\text{dom}(\text{est}(R)) \epsilon \text{ union est}(R)}$$

$$\begin{array}{ccc} \xleftarrow{\text{dom}(\text{est}(R)) \sqsubseteq \text{est}(R) \text{ est}(R)^\circ} & \boxed{\text{est}(R) \text{ est}(R)^\circ \epsilon \text{ union est}(R)} & \xleftarrow{\text{est}(R)^\circ \sqsubseteq \epsilon, \epsilon \epsilon \text{ union} \sqsubseteq \epsilon} \boxed{\text{est}(R) \epsilon \text{est}(R)} \\ & \boxed{\text{est}(R) R^\circ} & \boxed{\text{est}(R) R^\circ} \\ & & \text{UP of est} \end{array}$$

§10.3. Shortest paths on a cylinder, on lists

(10.3a)

	definition	type	note
1	$F(A,X)=A+A \times X$	$\mathcal{A} \rightarrow \mathcal{A}$	The base functor: one square of the new column, alone or in front of the path so far.
2	$L = \text{list}^+$	$\mathcal{A} \rightarrow \mathcal{A}$	A path is a non-empty list of squares, one per column crossed.
3	α the initial algebra	$F(\mu F) \rightarrow \mu F$	$\alpha(5)=[5]$ and $\alpha(1,[5])=[1,5]$: a path is started by one square, or extended by one.
4	N the n-tuple relator	$\mathcal{A} \rightarrow \mathcal{A}$	One component per row of a column, so the fold carries n answers at once.
5	$R \triangleq \text{sum} \leq \text{sum}^\circ$	$[\mathbb{N}] \rightarrow [\mathbb{N}]$	The cost of a path, which the cheapest minimises.
6	setify	$A[n] \rightarrow PA$	$\text{setify}(1,2,3,4)=\{1,2,3,4\}$ — which row a component came from is forgotten.
7	moves	$A[n] \rightarrow P(A[n])$	$\text{moves}(x)=\{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.
8	trans	$P(A[n]) \rightarrow (P\ A)[n]$	$\text{trans}\{(a,b,c), (x,y,z)\} = (\{a,x\}, \{b,y\}, \{c,z\})$ — component k of the result is the set of the k-th components.
9	zip	$F(a[n]) \rightarrow (F\ a)[n]$	$\text{zip}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$.
10	$\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$	$F(PA) \rightarrow PFA$	$\text{cp}(1, \{[5], [6], [8]\}) = ((1, [5]), (1, [6]), (1, [8]))$.
11	$\text{gen} \triangleq F(1, \text{moves trans } N(\text{union}))$ $\text{zip } N(\text{cp}(F(A, -))\ E(\alpha))$	$F(NA, N(P(TA))) \rightarrow N(P(TA))$	$\text{gen}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -))\ E(\alpha))$.
12	$\text{paths} \triangleq (\text{gen})\ \text{setify}\ \text{union}$	$T(NA) \rightarrow P(TA)$	$\text{paths}[(1,2,3,4), (5,6,7,8)]$ is the union of $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -))\ E(\alpha))$'s four sets: 12 paths, 3 from each entry row.
13	the specification $\text{paths est}(R)$	$[\mathbb{N}[n]] \rightarrow [\mathbb{N}]$	A cheapest path from the entry side to the exit side.

§10.3.1. $N(E(L\ A))$

(10.3.1a)

```

1  5
2  6      [(1,2,3,4), (5,6,7,8)] : L N Nat
3  7
4  8

```

(10.3.1b)

```

gen  $\triangleq$  F(1, moves trans N(union)) zip N(cp(F(A, -)) E( $\alpha$ ))
(gen)[(5,6,7,8)] = ({[5]}, {[6]}, {[7]}, {[8]})

gen((1,2,3,4), ({[5]}, {[6]}, {[7]}, {[8]})) =
( {[1,5]}, {[1,6]}, {[1,8]}},
  {[2,5]}, {[2,6]}, {[2,7]}},
  {[3,6]}, {[3,7]}, {[3,8]}},
  {[4,5]}, {[4,7]}, {[4,8]}} )

```

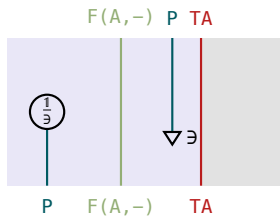
(10.3.1c)

1	N	the 4-tuple, one component per row of the entry column
2	E	the set of paths that can start in that row
3	L	a path is the non-empty list of the squares it crosses

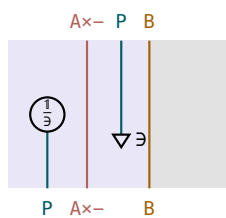
row 1's set has no $[1,7]$ — from row 1 only rows 4, 1 and 2 are reachable — and it does have $[1,8]$, because the cylinder glues the bottom row to the top. That is $\text{moves trans } N(\text{union})$ of $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$: component k collects the paths of rows $k-1, k, k+1$.

§10.3.2. The cross product $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$

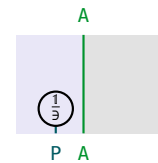
(10.3.2a)



cp on all of $F(A, E(L A))$



the $A \times -$ summand: $\exists$ picks one path, $1\% \exists$ collects the A summand, no E to distribute: $1\% \exists$ alone, $a \mapsto \{a\}$



(10.3.2b)

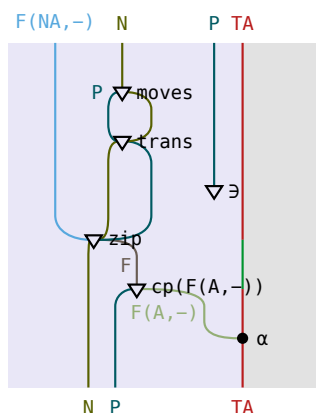
type	note
1 $F(A, -) = A + A \times -$	one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$
2 $\exists : E(L A) \rightarrow L A$	the $\exists$ inside $F(1, \exists)$: a set of paths, one of them — $L A$ is one path, A one square
3 $F(1, \exists) = 1 + 1 \times \exists$	$: A + A \times E(L A) \rightarrow A + A \times L A$, the relator acting on each summand
4 $\exists : E(A + A \times L A) \rightarrow A + A \times L A$	the $\exists$ under the bar: a set of cells, one of them — a different $\exists$
5 $\text{cp} : A + A \times E(L A) \rightarrow E(A + A \times L A)$	$\frac{R}{\exists}$ turns $R : X \rightarrow Y$ into $X \rightarrow E Y$
6 $\text{cp} = \frac{1}{\exists} E(F(1, \exists))$	what the picture draws: the unit makes the outer E , leaving one $\exists$ — $\frac{R}{\exists} = \frac{1}{\exists} E(R)$

(10.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$ the right summand
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$ 1 keeps the square 1, $\exists$ picks one path
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$ one output per path in w
 $\text{cp}(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$ the three of them, collected

§10.3.3. $\text{gen} = F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp } P(\alpha))$

(10.3.3a)

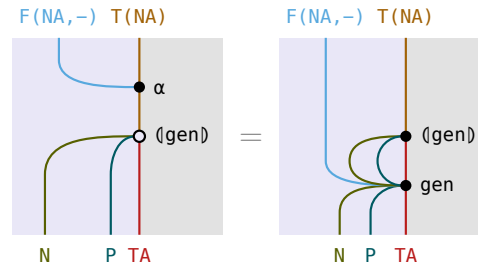


(10.3.3b)

$\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$
 $u = ((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) : F(N A, N(E(L A)))$
 $F(1, \text{moves trans } N(\text{union}))$ 1 keeps the column, the path SETS move
 $\text{moves}(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$
 $= (\{[6]\}, \{[7]\}, \{[8]\}, \{[5]\}),$
 $(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}),$
 $(\{[8]\}, \{[5]\}, \{[6]\}, \{[7]\})$: $E(N(E(L A)))$ down, unmoved, up
 $\text{trans}(\text{that})$
 $= (\{[6]\}, \{[5]\}, \{[8]\}, \{[7]\}, \{[6]\}, \{[5]\}),$
 $\{[8]\}, \{[7]\}, \{[6]\}, \{[5]\}, \{[8]\}, \{[7]\})$: $N(E(E(L A)))$ row k gets rows k-1, k, k+1
 $N(\text{union})(\text{that})$
 $= (\{[5], [6], [8]\}, \{[5], [6], [7]\},$
 $\{[6], [7], [8]\}, \{[5], [7], [8]\})$: $N(E(L A))$ every path into row k, none dropped
 $\text{zip}(\text{that})$
 $= ((1, \{[5], [6], [8]\}), (2, \{[5], [6], [7]\}),$
 $(3, \{[6], [7], [8]\}), (4, \{[5], [7], [8]\}))$: $N(F(A, E(L A)))$ each row: its square, and the paths it may be put in front of
 $N(\text{cp } P(\alpha))(\text{that})$
 $= (\{[1, 5], [1, 6], [1, 8]\}, \{[2, 5], [2, 6], [2, 7]\},$
 $\{[3, 6], [3, 7], [3, 8]\}, \{[4, 5], [4, 7], [4, 8]\})$: $N(E(L A))$ cp pairs the square with each path, α prefixes it: $\alpha(1, [5]) = [1, 5]$

§10.3.4. gen is an F-algebra; $(\text{gen}) : \alpha(\text{gen}) = F(1, (\text{gen})) \text{gen}$

(10.3.4a)

 α puts the column back on the list, then the fold reads all of itthe fold reads the rest under F , then one gen puts the column in front

$$xs = [(1, 2, 3, 4), (5, 6, 7, 8)] = \alpha((1, 2, 3, 4), [(5, 6, 7, 8)]) : L(N A)$$

(10.3.4b)

$$(\text{gen})(xs) = \text{gen}((1, 2, 3, 4), (\text{gen})[(5, 6, 7, 8)])$$

both halves are in $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$: the tail folds to $(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$, one path per row and each of them one square long, and gen on it is $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$'s walk.

§10.3.5. $Q : F(N A, N(L A)) \rightarrow N(L A)$, an $F(N A, -)$ -algebra: $Q = F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$

(10.3.5a)

$$Q : F(N A, N(L A)) \rightarrow N(L A)$$

the fold's algebra: a new column, one path per row

$u = ((1, 2, 3, 4), ([5], [6], [7], [8])) : F(N A, N(L A))$
 $F(1, \text{moves trans } N(\text{est}(R)))$ 1 keeps the column, the paths move
 $\text{moves}([5], [6], [7], [8])$
 $= ([6], [7], [8], [5]),$
 $([5], [6], [7], [8]),$
 $([8], [5], [6], [7])$: $E(N(L A))$ down, unmoved, up
 $\text{trans}(\text{that})$
 $= (\{[6], [5], [8]\}, \{[7], [6], [5]\},$
 $\{[8], [7], [6]\}, \{[5], [8], [7]\})$: $N(E(L A))$ row k gets rows k-1, k, k+1
 $N(\text{est}(R))(\text{that})$
 $= ([5], [5], [6], [5])$: $N(L A)$ the cheapest into each row – chosen BEFORE the new square
 $\text{zip}(\text{that})$
 $= ((1, [5]), (2, [5]), (3, [6]), (4, [5]))$: $N(F(A, L A))$ each row: its square, and the one predecessor that survived
 $N(\alpha)(\text{that})$
 $= ([1, 5], [2, 5], [3, 6], [4, 5])$: $N(L A)$ $\alpha(1, [5]) = [1, 5]$

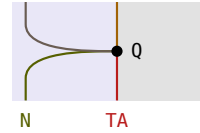
$\text{lax}(\text{moves}) \wedge \text{lax}(\text{trans}) \wedge \text{lax}(\text{zip}) \wedge F(1, \text{moves trans } N(\text{est}(R))) = [\text{l}, \text{pm } r] \wedge \text{zip} =$ (10.3.5b)
 $[N(\text{inl}), \text{zip}' N(\text{inr})] \wedge \text{inl } \alpha = \text{wrap} \wedge \text{inr } \alpha = \text{cons} \Rightarrow Q = [N(\text{wrap}), \text{pm } \text{zip}' N(\text{cons})]$
 at the last column Q starts one path per row, and at each earlier one it puts each square in front of the
 cheapest of the three kept paths it can step to — the algebra $\text{lax}(\text{moves}) \wedge \text{lax}(\text{trans}) \wedge \text{lax}(\text{zip}) \wedge$
 $\text{lax}(\text{setify}) \wedge R R \sqsubseteq R \wedge F(1, N(\text{est}(R))) Q \sqsubseteq \text{gen } N(\text{est}(R))$
 $\Rightarrow (Q) \text{ setify } \text{est}(R) \sqsubseteq \text{paths } \text{est}(R)$'s last step folds

circuit — the fork is $F(NA, N(LA)) = NA + NA \times N(LA)$

Hinze–Marsden

$F(N A) (N (T A)) \text{---} \boxed{Q} \text{---} N(T A)$

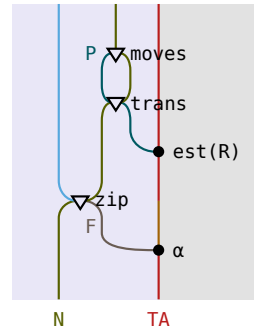
$F(N A, N (T A))$



$FN(T A) \text{---} \boxed{F(1, \text{moves trans } N(\text{est}(R)))} \text{---} FN(T A) \text{---} \boxed{\text{zip}} \text{---} N(F A (T A)) \text{---} \boxed{N(\alpha)} \text{---} N(T A)$

$=$ the fusion condition $\text{lax}(\text{moves}) \wedge \text{lax}(\text{trans}) \wedge \text{lax}(\text{zip}) \wedge R R \sqsubseteq R \wedge F(1, \text{est}(R)) \alpha \sqsubseteq$
 $\text{cp}(F(A, -)) E(\alpha) \text{est}(R)$
 $\Rightarrow F(1, N(\text{est}(R))) Q \sqsubseteq \text{gen } N(\text{est}(R))$ read as a definition,

$F(NA, -) N TA$



$=$ $F(N A) (N (T A)) \text{---} \boxed{\begin{array}{c} \text{l} \text{---} N(p_0) \text{---} N(\text{wrap}) \\ \text{r} \text{---} q \text{---} \text{pm} \text{---} q \text{---} \text{zip}' \text{---} N(q_0) \text{---} N(\text{cons}) \end{array}} \text{---} N(T A)$
 $\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]$

$$\text{lax(moves)} \wedge \text{lax(trans)} \wedge \text{lax(zip)} \wedge \text{lax(setify)} \wedge R \ R \sqsubseteq R \wedge F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen} \quad (10.3.5c)$$

$$N(\text{est}(R)) \Rightarrow \langle Q \rangle \text{ setify } \text{est}(R) \sqsubseteq \text{paths } \text{est}(R)$$

shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in $O(n \times m)$ time; Q is $\text{lax(moves)} \wedge \text{lax(trans)} \wedge \text{lax(zip)} \wedge$

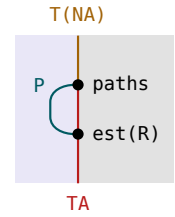
$$F(1, \text{moves trans } N(\text{est}(R))) = [l, pm \ r] \wedge \text{zip} = [N(\text{inl}), \text{zip}' \ N(\text{inr})] \wedge \text{inl } \alpha = \text{wrap} \wedge \text{inr } \alpha = \text{cons}$$

$$\Rightarrow Q = [N(\text{wrap}), pm \ \text{zip}' \ N(\text{cons})] \text{ 's algebra.}$$

circuit

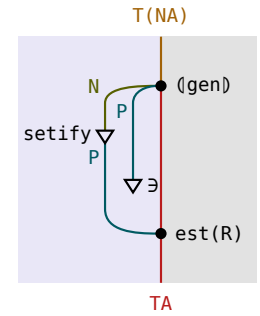
Hinze–Marsden

$$T(N \ A) \text{---} \boxed{\text{paths}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$



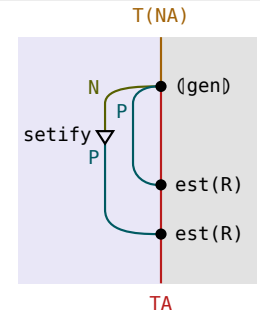
$$= T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (P \ (T \ A)))]} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{\text{setify}} \text{---} PP(T \ A) \text{---} \boxed{\text{union}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

$\text{paths} \triangleq \langle \text{gen} \rangle \text{ setify union at paths}$



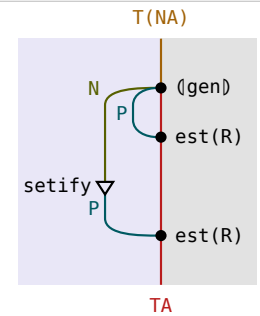
$$\sqsubseteq T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (P \ (T \ A)))]} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{\text{setify}} \text{---} PP(T \ A) \text{---} \boxed{P(\text{est}(R))} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

$P(\text{est}(R)) \ \text{est}(R) \sqsubseteq \text{union } \text{est}(R)$ — a minimum in each set, then a minimum of those; R transitive



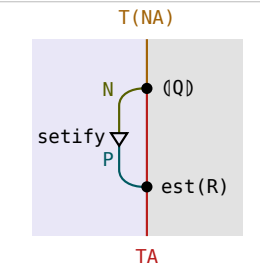
$$\sqsubseteq T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (P \ (T \ A)))]} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T \ A) \text{---} \boxed{\text{setify}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

$\text{setify lax natural}$



$$\sqsubseteq T(N \ A) \text{---} \boxed{[F(N \ A) \ (N \ (T \ A))]} \text{---} \boxed{Q} \text{---} N(T \ A) \text{---} \boxed{\text{setify}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

fusion at $\text{lax(moves)} \wedge \text{lax(trans)} \wedge \text{lax(zip)} \wedge R \ R \sqsubseteq R \wedge F(1, \text{est}(R)) \ \alpha \sqsubseteq \text{cp}(F(A, -))$
 $E(\alpha) \ \text{est}(R)$
 $\Rightarrow F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen } N(\text{est}(R))$



$$\text{lax}(\text{moves}) \wedge \text{lax}(\text{trans}) \wedge \text{lax}(\text{zip}) \wedge R \subseteq R \wedge F(1, \text{est}(R)) \alpha \subseteq \text{cp}(F(A, -)) E(\alpha) \text{est}(R) \Rightarrow (10.3.5d)$$

$$F(1, N(\text{est}(R))) Q \subseteq \text{gen } N(\text{est}(R))$$

$$\text{fusion: the condition for fusion in } \text{lax}(\text{moves}) \wedge \text{lax}(\text{trans}) \wedge \text{lax}(\text{zip}) \wedge \text{lax}(\text{setify}) \wedge R \subseteq R \wedge F(1, N(\text{est}(R))) Q \subseteq \text{gen } N(\text{est}(R))$$

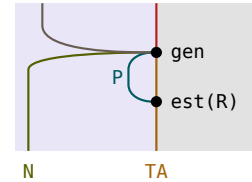
$$\Rightarrow (Q) \text{ setify } \text{est}(R) \subseteq \text{paths } \text{est}(R) \text{ 's last step, used to derive a definition of } Q.$$

circuit

$$F(N(A) (N(P(T A)))) \xrightarrow{\text{gen}} NP(T A) \xrightarrow{N(\text{est}(R))} N(T A)$$

Hinze–Marsden

$$F(N(A), N(P(T A)))$$

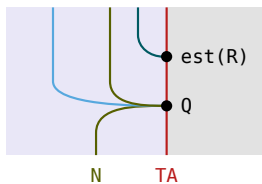


N

TA

$$\begin{aligned} & \text{FNP}(T A) \xrightarrow{F(1, N(\text{est}(R)))} FN(T A) \xrightarrow{Q} N(T A) \\ & \text{(7.13), then zip, trans, moves lax natural} \end{aligned}$$

$$F(NA, -) \quad N \quad P \quad TA$$



N

TA

$$(7.13) \text{ is } F(1, \text{est}(R)) \alpha \subseteq \text{cp } P(\alpha) \text{est}(R), \text{ Theorem 7.1 Map}(f) \wedge 1 \subseteq R$$

$$\Rightarrow f \circ F(R) f \subseteq R \Leftrightarrow F(\text{est}(R)) f \subseteq \frac{F(\exists) f}{\exists} \text{est}(R) \text{ at the map } \alpha \text{ with } \frac{F(1, \exists) \alpha}{\exists} = \text{cp } P(\alpha): \text{ extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.}$$